

Theorem 5.2.6. (Hajós 1961)

Let G be a graph and $k \in \mathbb{N}$. Then $\chi(G) \geq k$ if and only if G has a k -constructible subgraph.

Proof. Let G be a graph with $\chi(G) \geq k$; we show that G has a k -constructible subgraph. Suppose not; then $k \geq 3$. Adding some edges if necessary, let us make G edge-maximal with the property that none of its subgraphs is k -constructible. Now G is not a complete r -partite graph for any r : for then $\chi(G) \geq k$ would imply $r \geq k$, and G would contain the k -constructible graph K^k .

Since G is not a complete multipartite graph, non-adjacency is not an equivalence relation on $V(G)$. So there are vertices y_1, x, y_2 such that $y_1x, xy_2 \notin E(G)$ but $y_1y_2 \in E(G)$. Since G is edge-maximal without a k -constructible subgraph, each edge xy_i lies in some k -constructible subgraph H_i of $G + xy_i$ ($i = 1, 2$).

Let H'_2 be an isomorphic copy of H_2 that contains x and $H_2 - H_1$ but is otherwise disjoint from G , together with an isomorphism $v \mapsto v'$ from H_2 to H'_2 that fixes $H_2 \cap H'_2$ pointwise. Then $H_1 \cap H'_2 = \{x\}$, so

$$H := (H_1 \cup H'_2) - xy_1 - xy'_2 + y_1y'_2$$

is k -constructible by (iii). One vertex at a time, let us identify in H each vertex $v' \in H'_2 - G$ with its partner v ; since vv' is never an edge of H , each of these identifications amounts to a construction step of type (ii). Eventually, we obtain the graph

$$(H_1 \cup H_2) - xy_1 - xy_2 + y_1y_2 \subseteq G;$$

this is the desired k -constructible subgraph of G . $\square$

Does Hajós's theorem solve our Kuratowski-type problem for highly chromatic graphs, which was to find a class of graphs of chromatic number at least k with the property that every such graph has a subgraph in this class? Formally it does, albeit with an infinite characterizing class: the class of k -constructible graphs, which contains $\mathcal{X}_k$. Unlike Kuratowski's characterization of planar graphs, however, this does not – at least not obviously – make Hajós's theorem a good characterization of the graphs of chromatic number $< k$: as one can show, proving that a given k -constructible graph is indeed k -constructible is just as hard as proving that a graph of chromatic number $\geq k$ does indeed need at least k colours. See the notes for details.

5.3 Colouring edges

Clearly, every graph G satisfies $\chi'(G) \geq \Delta(G)$. For bipartite graphs, we have equality here:

[5.4.5]

Proposition 5.3.1. (König 1916)

Every bipartite graph G satisfies $\chi'(G) = \Delta(G)$.

(1.6.1)

 Δ, xy α -edge α, β

Proof. We apply induction on $\|G\|$. For $\|G\| = 0$ the assertion holds. Now assume that $\|G\| \geq 1$, and that the assertion holds for graphs with fewer edges. Let $\Delta := \Delta(G)$, pick an edge $xy \in G$, and choose a Δ -edge-colouring of $G - xy$ by the induction hypothesis. Let us refer to the edges coloured α as α -edges, etc.

In $G - xy$, each of x and y is incident with at most $\Delta - 1$ edges. Hence there are $\alpha, \beta \in \{1, \dots, \Delta\}$ such that x is not incident with an α -edge and y is not incident with a β -edge. If $\alpha = \beta$, we can colour the edge xy with this colour and are done; so we may assume that $\alpha \neq \beta$, and that x is incident with a β -edge.

Let us extend this edge to a maximal walk W from x whose edges are coloured β and α alternately. Since no such walk contains a vertex twice (why not?), W exists and is a path. Moreover, W does not contain y : if it did, it would end in y on an α -edge (by the choice of β) and thus have even length, so $W + xy$ would be an odd cycle in G (cf. Proposition 1.6.1). We now recolour all the edges on W , swapping α with β . By the choice of α and the maximality of W , adjacent edges of $G - xy$ are still coloured differently. We have thus found a Δ -edge-colouring of $G - xy$ in which neither x nor y is incident with a β -edge. Colouring xy with β , we extend this colouring to a Δ -edge-colouring of G . $\square$

Theorem 5.3.2. (Vizing 1964)

Every graph G satisfies

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

 V, E Δ

colouring

Proof. We prove the second inequality by induction on $\|G\|$. For $\|G\| = 0$ it is trivial. For the induction step let $G = (V, E)$ with $\Delta := \Delta(G) > 0$ be given, and assume that the assertion holds for graphs with fewer edges. Instead of ' $(\Delta + 1)$ -edge-colouring' let us just say 'colouring'.

missing

 α/β -path

For every edge $e \in G$ there exists a colouring of $G - e$, by the induction hypothesis. In such a colouring, the edges at a given vertex v use at most $d(v) \leq \Delta$ colours, so some colour $\beta \in \{1, \dots, \Delta + 1\}$ is *missing at v* . For any other colour α , there is a unique maximal walk (possibly trivial) starting at v , whose edges are coloured alternately α and β . This walk is a path; we call it the α/β -path from v .

Suppose that G has no colouring. Then the following holds:

Given $xy \in E$, and any colouring of $G - xy$ in which the colour α is missing at x and the colour β is missing at y ,
the α/β -path from y ends in x . (*)

Otherwise we could interchange the colours α and β along this path and colour xy with α , obtaining a colouring of G (contradiction).

Let $xy_0 \in G$ be an edge. By induction, $G_0 := G - xy_0$ has a colouring c_0 . Let α be a colour missing at x in this colouring. Further, let $y_0, \dots, y_k$ be a maximal sequence of distinct neighbours of x in G such that $c_0(xy_{i+1})$ is missing in c_0 at y_i for every $i < k$. For each of the graphs $G_i := G - xy_i$ we define a colouring c_i , setting

$$c_i(e) := \begin{cases} c_0(xy_{j+1}) & \text{for } e = xy_j \text{ with } j \in \{0, \dots, i-1\} \\ c_0(e) & \text{otherwise;} \end{cases}$$

note that in each of these colourings the same colours are missing at x as in c_0 .

Now let β be a colour missing at y_k in c_0 . By (*), the α/β -path P from y_k in G_k (with respect to c_k) ends in x , with an edge yx coloured β since α is missing at x . Since y cannot serve as y_{k+1} , by the maximality of the sequence $y_0, \dots, y_k$, we thus have $y = y_i$ for some $0 \leq i < k$ (Fig. 5.3.1). By definition of c_k , therefore, $\beta = c_k(xy_i) = c_0(xy_{i+1})$. By the choice of y_{i+1} this means that β was missing at y_i in c_0 , and hence also in c_i . Now the α/β -path P' from y_i in G_i with respect to c_i starts with y_iPy_k , since the edges of P are coloured the same in c_i as in c_k . But in c_0 , and hence in c_i , there is no edge at y_k coloured β . Therefore P' ends in y_k , contradicting (*). $\square$

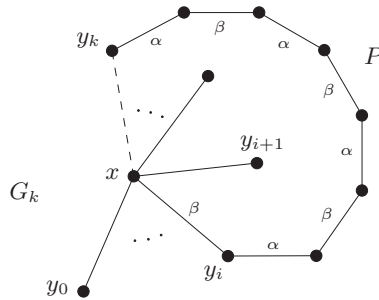


Fig. 5.3.1. The α/β -path P in $G_k = G - xy_k$

Vizing's theorem divides the finite graphs into two classes according to their chromatic index; graphs satisfying $\chi' = \Delta$ are called (imaginatively) *class 1*, those with $\chi' = \Delta + 1$ are *class 2*. There is no good characterization theorem that enables us to tell these classes apart, because no easily checkable 'certificate' is known for a graph to be class 2.

Regular graphs of large even order and large degree are class 1:

Theorem 5.3.3. (Csaba, Kühn, Lo, Osthus, Treglown 2016)

There exists an $n_0 \in \mathbb{N}$ such that, for all even $n \geq n_0$ and $d \geq n/2$, every d -regular graph G of order n satisfies $\chi'(G) = \Delta(G)$.

5.4 List colouring

In this section, we take a look at a relatively recent generalization of the concepts of colouring studied so far. This generalization may seem a little far-fetched at first glance, but it turns out to supply a fundamental link between the classical (vertex and edge) chromatic numbers of a graph and its other invariants.

Suppose we are given a graph $G = (V, E)$, and for each vertex of G a list of colours permitted at that particular vertex: when can we colour G (in the usual sense) so that each vertex receives a colour from its list? More formally, let $(S_v)_{v \in V}$ be a family of sets. We call a vertex colouring c of G with $c(v) \in S_v$ for all $v \in V$ a colouring *from the lists* S_v . The graph G is called *k-list-colourable*, or *k-choosable*, if, for every family $(S_v)_{v \in V}$ with $|S_v| = k$ for all v , there is a vertex colouring of G from the lists S_v . The least integer k for which G is *k-choosable* is the *list-chromatic number*, or *choice number* $\text{ch}(G)$ of G .

List-colourings of edges are defined analogously. The least integer k such that G has an edge colouring from any family of lists of size k is the *list-chromatic index* $\text{ch}'(G)$ of G ; formally, we just set $\text{ch}'(G) := \text{ch}(L(G))$, where $L(G)$ is the line graph of G .

In principle, showing that a given graph is *k-choosable* is more difficult than proving it to be *k-colourable*: the latter is just the special case of the former where all lists are equal to $\{1, \dots, k\}$. Thus,

$$\text{ch}(G) \geq \chi(G) \quad \text{and} \quad \text{ch}'(G) \geq \chi'(G)$$

for all graphs G .

In spite of these inequalities, many of the known upper bounds for the chromatic number have turned out to be valid for the choice number, too. Examples for this phenomenon include Brooks's theorem and Proposition 5.2.2; in particular, graphs of large choice number still have subgraphs of large minimum degree. On the other hand, it is easy to construct graphs for which the two invariants are wide apart (Exercise 30). Taken together, these two facts indicate a little how far those general upper bounds on the chromatic number may be from the truth.

The following theorem shows that, in terms of its relationship to other graph invariants, the choice number differs fundamentally from the chromatic number. As mentioned before, there are 2-chromatic graphs of arbitrarily large minimum degree, e.g. the graphs $K_{n,n}$. The choice number, however, will be forced up by large values of invariants like δ , ε or κ :

Theorem 5.4.1. (Alon 1993)

There exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, given any integer k , all graphs G with average degree $d(G) \geq f(k)$ satisfy $\text{ch}(G) \geq k$.